

Highlights

A Reproducible Workflow for Whole-Slide Cancer Detection

Bruno Henrique Nunes

- End-to-end pathology workflow with auditable intermediate artifacts.
- DIAGSET evaluation with patient-level splits and train-derived stain state.
- Metric-specific normalization findings across Dice, AUROC, and patch AvAcc.
- Negative-patient false positives remain the main limitation.

A Reproducible Workflow for Whole-Slide Cancer Detection^{*}

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ABSTRACT

Whole-slide histopathology experiments are vulnerable to failures that occur before model fitting, including annotation ambiguity, contaminated patches, patient-level leakage, color variability, incompatible intermediate artifacts, and validation choices that leak into the test set. This work presents a reproducibility-oriented workflow for cancer segmentation in digital pathology and evaluates it on DIAGSET, a public prostate histopathology dataset. The workflow converts slide and annotation pairs into HDF5-backed patch datasets, performs manual-review-guided graph cleaning, generates patient-disjoint training, validation, and test splits, applies stain normalization at runtime from training-derived state, optionally reduces only the training split by embedding-based smart sampling, trains a fixed registry of segmentation models, optimizes a two-stream ensemble on validation patients, and evaluates the frozen recipe on held-out test patients. In the DIAGSET experiment, 419 valid image–annotation cases yielded 1,186,778 retained patches across 393 patients; the held-out test set contained 79 patients and 387,325 patches. No stain-normalization strategy was uniformly best across evaluation criteria. The non-normalized setting obtained the highest pixel-level micro Dice (0.8415; 95% CI 0.7767–0.8987) and patient-level macro Dice (0.7540; 95% CI 0.6710–0.8356), Ruifrok obtained the highest AUROC point estimate (0.9579) and positive-patient Dice (0.7985; 95% CI 0.7189–0.8688), and Macenko obtained the highest DIAGSET-style patch balanced accuracy before patient suppression (AvAcc 0.9009). However, all settings still produced false positives in at least 11 of 41 negative test patients. The results support the workflow as an auditable experimental protocol, while showing that clinical interpretation remains limited by false-positive control and by evaluation on a single dataset.

1. Introduction

Deep-learning pipelines for whole-slide histopathology are shaped as much by data preparation as by the final neural architecture. A model can appear effective while relying on ambiguous annotations, repeated patient information across splits, contaminated training patches, preprocessing state derived from evaluation data, or test-time thresholds adjusted after inspecting held-out performance. These risks are especially consequential in computational pathology, where gigapixel images are converted into many smaller patches and where patient-level correlation can make naive patch-level evaluation overly optimistic [21, 5].

This paper reports a workflow designed to make those experimental decisions explicit and auditable. Rather than treating cancer detection as a single training script, the repository separates slide ingestion, patch extraction, review-oriented cleaning, patient-level split generation, stain-normalization state construction, sanity checking, training-sample selection, model fitting, validation-only ensemble optimization, and frozen test inference. The resulting artifact chain is intended to reduce silent reuse of stale data, preserve image–mask alignment, and make final results traceable to specific configurations and intermediate decisions [28, 19].

The empirical evaluation uses DIAGSET, a public prostate histopathology dataset with image and annotation

material suitable for cancer versus non-cancer analysis [13]. The main experimental question is not whether one color-normalization method is universally superior, but whether a complete, provenance-preserving workflow can compare normalization strategies under a fixed training, validation, and test protocol. Five runtime settings are evaluated: no color normalization, Reinhard, Ruifrok, Macenko, and Vahadane [20, 22, 14, 26].

The results show that the preferred normalization depends on the evaluation lens. Pixel- and patient-level Dice favor the non-normalized setting, AUROC and positive-patient Dice favor Ruifrok, and DIAGSET-style patch AvAcc favors Macenko. This disagreement is not a nuisance result; it is the central interpretation. In segmentation tasks with both positive and negative patients, micro Dice, macro Dice, AUROC, patch-recognition metrics, and negative-patient false-positive control answer different scientific questions [23, 8, 24]. The paper therefore emphasizes metric-specific interpretation and conservative scope rather than a single winning configuration.

2. Related Work

Whole-slide analysis often starts with quality control, tissue detection, and patch extraction because WSIs are too large to process directly and can contain artifacts that degrade downstream learning [27, 17, 12]. The present workflow follows this tradition but makes patch identities, HDF5 row alignment, and provenance checks explicit, so that downstream learning consumes a stable dataset contract rather than an informal folder of image files [10, 28].

Color variation is another persistent issue in histopathology. Stain normalization and stain augmentation

^{*}This manuscript reports the DIAGSET experiment from the repository and thesis artifacts. Claims are restricted to the validated workflow and available DIAGSET outputs.

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have been studied as ways to reduce scanner- and protocol-dependent appearance shifts, but their effect can be dataset- and metric-dependent [25]. The workflow therefore stores training-derived normalization state and applies the selected normalization at runtime, allowing Reinhard, Ruifrok, Macenko, Vahadane, and non-normalized inference to share the same patient split. For Vahadane, this paper reports the configured fixed-source approximation, not exact per-patch source fitting.

The final prediction stage is related to ensemble selection and test-time augmentation, but its exact two-stream region-of-interest gating objective is repository-specific [6, 4, 16]. Validation patients are used to optimize ensemble weights and thresholds, while the test set is consumed only after the recipe is frozen. This split between validation-only recipe design and held-out inference is central to the paper's claim of reproducible evaluation.

3. Materials and Methods

3.1. Workflow Overview

The workflow is organized as ten numbered stages connected by explicit files, database rows, and metadata sidecars. Figure 1 condenses those stages into the main artifact path from slide ingestion to frozen test inference. Early stages build the dataset contract; middle stages curate, split, and sample the data; late stages train single models, optimize a validation-only ensemble recipe, and evaluate that recipe on held-out test patients. Root scripts orchestrate each stage, while reusable logic lives in domain-specific helper modules.

3.2. Dataset and Curation

DIAGSET was used as the experimental dataset [13]. The ingestion stage registered 429 .ndpi images and 429 .ndpa annotations, of which 419 were paired and validated through converted GeoJSON annotations. Ten images were rejected or failed before final extraction. Patches were extracted with a 224-pixel window, 112-pixel stride, annotation-overlap threshold of 1.0, and minimum tissue-content threshold of 0.3.

Cancer-patch curation used a small manually reviewed pilot set and a larger labeled master set to tune a graph-based background-contamination rule. The pilot set contained 100 reviewed samples, with 94 approved and 6 rejected. The master set contained 385 samples, with 367 approved and 18 rejected. The selected graph-cleaning configuration used a background-intensity threshold of 167, Felzenszwalb scale parameter $k = 140$, minimum segment size 97, a 3-pixel erosion radius, and contamination threshold $\tau = 0.19$ [9]. Applying this rule rejected 74,047 cancer patches and retained 1,186,778 patches for downstream analysis. Non-cancer patches were not rejected by this graph-cleaning step because the rule was targeted at positive regions contaminated by background or annotation mismatch.

3.3. Patient-Level Splitting and Runtime Normalization

After cleaning, the retained DIAGSET patches were split at the stored patient-identifier level. The final dataset contained 393 patients: 236 for training, 78 for validation, and 79 for testing. The test partition contained 387,325 patches from 38 positive and 41 negative patients. This split policy is intended to prevent patches associated with the same stored patient identifier from appearing in more than one partition, while preserving the limitation that the repository does not independently prove biological patient identity beyond the identifiers available at ingestion.

Stain normalization is selected downstream through `RUNTIME_NORMALIZATION_METHOD` rather than being baked into separate split assignments. Stage 4 constructs training-derived template and runtime-normalization state for Reinhard, Ruifrok, Macenko, and Vahadane, while the non-normalized condition is treated as a no-transform setting. This design allows all normalization strategies to reuse the same patient split and avoids deriving preprocessing state from validation or test patients. The Vahadane experiment used `RUNTIME_VAHADANE_BACKEND=fixed_source`, a fixed-source approximation to per-patch Vahadane normalization; the reported Vahadane results should therefore be interpreted under that operational contract.

3.4. Training and Checkpoint Selection

Smart sampling was applied only to the training split. Of 660,867 available training patches, 180,381 were retained, corresponding to 27.29% of the training rows. Validation and test rows were not compacted or filtered, so sampling changed the effective training set without changing final evaluation data. The sampling stage used Phikon-v2 embeddings and stored selected training rows in compact `TRAIN_SELECTED_COMPACT_DIR` storage for faster downstream reads [3, 18, 11].

Each normalization setting was evaluated with the same eight registered architecture-encoder pairs, producing 40 final model trainings. All final trainings used seed 24. The model registry fixed the architecture search space, while training used a weighted binary-cross-entropy plus Dice objective derived from the implemented loss family, inverse-frequency patch sampling, and validation-driven checkpoint selection [12, 11]. The loss was

$$\mathcal{L} = \alpha \mathcal{L}_{\text{BCE},fg} + \beta \mathcal{L}_{\text{Dice},bg} + \gamma \mathcal{L}_{\text{Dice},fg}, \quad (1)$$

with $\alpha = 0.125$, $\beta = 0.157$, and $\gamma = 0.173$ in the final protocol. Checkpoints were selected by validation pixel-level AUPRC, with AUROC and MCC tracked as auxiliary metrics [23, 15].

3.5. Two-Stream Ensemble Inference

For each normalization setting, the final inference recipe combined eight trained models into a `two_stream_spatial_gating` ensemble. Four models formed a semantic stream used to define a coarse region of interest, and four models formed a spatial stream used for finer segmentation inside

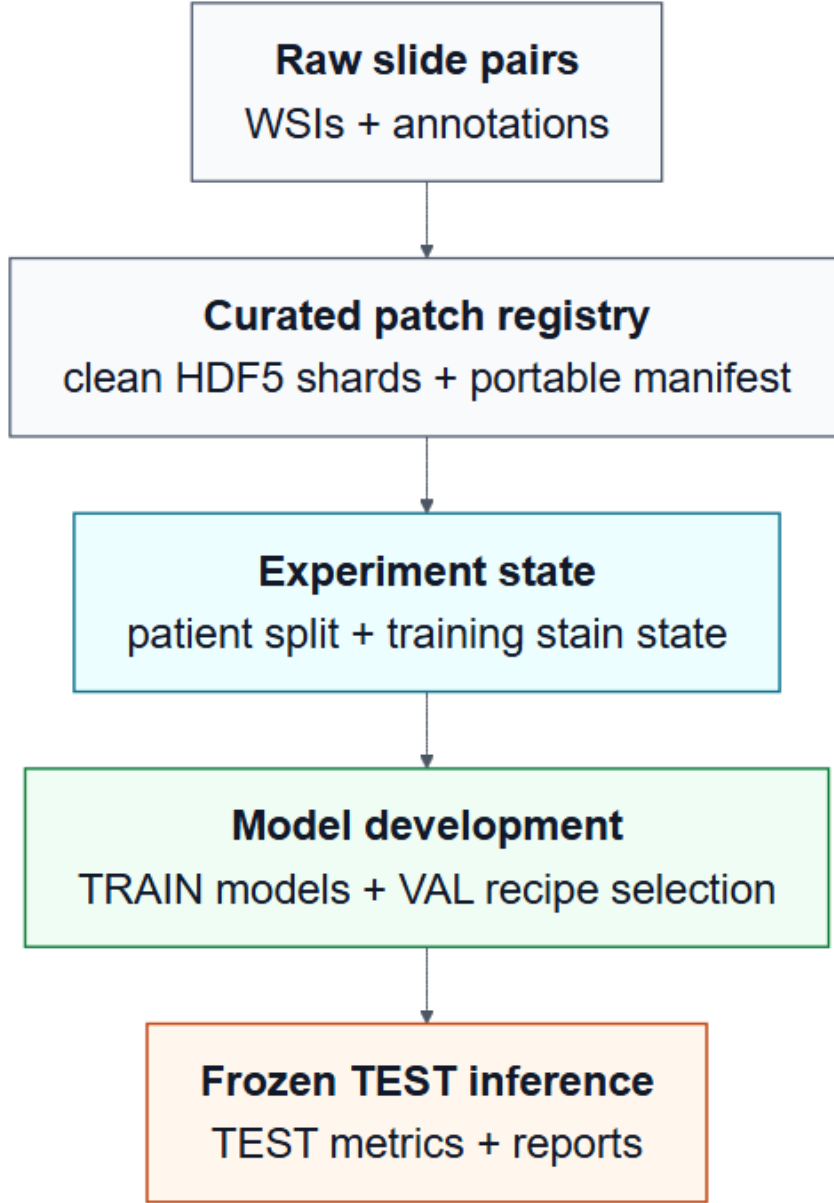


Figure 1: Condensed overview of the implemented workflow. The figure emphasizes the main reproducibility constraints: curated manifest-backed patches, patient-level split isolation, train-derived stain state, train-only sampling, validation-only recipe selection, and frozen held-out test inference.

that semantic support. The semantic stream optimized normalized model weights and an ROI threshold on validation patients, favoring positive-patient ROI recall while discouraging empty, overly permissive, and negative-patient ROIs. The spatial stream then optimized model weights under the fixed ROI by maximizing positive-patient AUPRC inside the ROI while penalizing probability spill outside the ROI and prediction mass on negative validation patients [1, 2, 23].

Let $P_{\text{sem}}(x)$ and $P_{\text{spa}}(x)$ be the semantic- and spatial-stream foreground probabilities at pixel x . The semantic ROI is generated by thresholding the downsampled semantic

prediction and upsampling it back to patch resolution,

$$R(x) = \text{Up}\left(\mathbf{1}_{\{\text{Down}(P_{\text{sem}}; s) > \tau_{\text{roi}}\}}\right)(x).$$

The final prediction is then

$$P_{\text{final}}(x) = P_{\text{spa}}(x)R(x), \quad \hat{Y}(x) = \mathbf{1}_{\{P_{\text{final}}(x) > \tau_{\text{dec}}\}}.$$

The final binary decision was calibrated on validation patients with a balanced Rule-6 grid search rather than a single threshold-only criterion. Each candidate jointly specified the decision threshold, a minimum connected-component area in pixels, a minimum connected-component area as a fraction of patch area, a minimum number of predicted-positive

patches per patient, and a minimum predicted-positive area fraction per patient. Before candidate scoring, per-patch connected components below the effective area threshold were removed and the resulting counts were accumulated at the patient level. The calibration objective combined Dice on positive calibration patients with a clean-negative indicator on negative calibration patients,

$$\text{Rule6}_{\text{cal}} = \frac{\sum_{j \in \mathcal{P}_+^{\text{cal}}} \text{Dice}_j + \sum_{j \in \mathcal{P}_-^{\text{cal}}} \mathbf{1}_{\{FP_j=0\}}}{|\mathcal{P}_+^{\text{cal}}| + |\mathcal{P}_-^{\text{cal}}|}.$$

Candidate selection also enforced baseline-relative positive-patient Dice and true-positive-rate guardrails, minimum micro Dice, minimum macro precision, minimum macro true-positive rate, and a target negative-clean rate. When these targets were infeasible, the recipe recorded the calibration status rather than silently treating the selected operating point as satisfying all constraints. The optimized ROI thresholds were tightly clustered near 0.15, while the final decision thresholds differed by normalization setting: 0.85 for no normalization, 0.90 for Reinhard, 0.65 for Ruifrok, 0.90 for Macenko, and 0.55 for Vahadane. These two thresholds summarize only part of the frozen recipe; the recipe also stores the validation-selected component, patch-count, patient-area post-processing parameters, and negative false-positive area diagnostics.

3.6. Metrics and Uncertainty

Final evaluation was performed only on the frozen test split. Pixel-level counts were accumulated after the validation-selected thresholding, connected-component filtering, and frozen patient-level suppression rules. Micro metrics were computed by pooling pixel counts across test patients before evaluating the metric, whereas macro metrics were computed per patient and then averaged. The reported summary includes Dice, IoU, accuracy, sensitivity, specificity, precision, false-positive and false-negative rates, AUROC, and a separate positive/negative-patient decomposition. Positive-patient Dice was computed only among patients with annotated lesion pixels. Negative-clean rate was defined as the proportion of negative patients with no positive prediction,

$$\text{NCR} = \frac{1}{|\mathcal{P}_-|} \sum_{j \in \mathcal{P}_-} \mathbf{1}_{\{FP_j=0\}},$$

where $\mathcal{P}_-$ is the set of negative patients and FP_j is the number of false-positive pixels for patient j .

Stage 10 also emits a DIAGSET-style patch-recognition summary without changing the segmentation decision path. For patch i , the prediction is converted to a binary patch label from the component-filtered segmentation mask,

$$\hat{z}_i = \mathbf{1}_{\{a(\hat{Y}_i) > \tau_{\text{patch}}\}},$$

where $a(\hat{Y}_i)$ is the predicted positive area fraction and the default $\tau_{\text{patch}} = 0$ marks a patch as cancer when any positive mask area remains after component filtering. Ground-truth

patch labels are taken from the manifest label produced by the extraction-overlap rule, with a binary-mask fallback when labels are absent from the batch. The patch report gives point estimates for Accuracy, Sensitivity, Specificity, and balanced average accuracy,

$$\text{AvAcc} = \frac{\text{Sensitivity} + \text{Specificity}}{2},$$

plus support and confusion counts. The primary DIAGSET-style comparison is recorded before patient-level suppression, and a second section reports the same patch metrics after the frozen patient-level suppression rule.

When confidence intervals were available, they were computed by nonparametric bootstrap with 10,000 resamples at the patient level [7]. This patient-wise procedure was used for micro metrics, macro metrics, and the positive/negative-patient decomposition. AUROC is reported as a point estimate only because the final artifacts preserved the information needed for histogram-based AUROC but did not preserve patient-indexed probability distributions required for a patient-level AUROC bootstrap.

4. Results

4.1. Curation, Split, and Sampling Outcomes

The preprocessing and curation stages produced a dataset large enough for model development while preserving patient-level evaluation. Starting from 429 registered image–annotation pairs, 419 cases passed pairing and annotation validation. Graph cleaning removed 74,047 positive patches and left 1,186,778 accepted patches. The resulting patient-level split preserved similar positive and negative patient counts: 115 positive and 121 negative patients in training, 43 positive and 35 negative patients in validation, and 38 positive and 41 negative patients in testing. Smart sampling then reduced only the training split from 660,867 to 180,381 patches.

These numbers are methodologically important because they define the evaluation contract. Curation occurred before patient splitting, smart sampling was restricted to training rows, checkpoints were selected on validation data, ensemble weights and thresholds were optimized on validation patients, and the test set remained held out until frozen inference.

4.2. Final Test Performance

Table 1 summarizes the final held-out test results for the five normalization strategies. The table is intentionally compact: it reports the metrics needed to interpret the main segmentation and patient-level findings without reproducing the full thesis inventory of per-model checkpoints and per-metric tables. Table 2 adds the DIAGSET-style patch-recognition summary for comparisons that require patch-level Accuracy or AvAcc rather than segmentation overlap.

The leading method depends on the metric. The non-normalized condition obtained the highest micro Dice and macro Dice, indicating the strongest overlap both after pooling all test pixels and after averaging patient-level results.

Table 1

Final DIAGSET test performance by runtime-normalization strategy. Dice intervals are 95% patient-bootstrap confidence intervals. AUROC is reported as a point estimate because patient-indexed AUROC bootstrap artifacts were not available. NCR denotes negative-clean rate, the fraction of negative patients with no positive prediction.

Normalization	Micro Dice	Macro Dice	AUROC	Positive-patient Dice	NCR
None	0.8415 [0.7767, 0.8987]	0.7540 [0.6710, 0.8356]	0.9508	0.7781 [0.6816, 0.8628]	0.7317 [0.5909, 0.8649]
Reinhard	0.8296 [0.7773, 0.8789]	0.7363 [0.6510, 0.8192]	0.9445	0.7413 [0.6385, 0.8327]	0.7317 [0.5909, 0.8649]
Ruifrok	0.8295 [0.7786, 0.8710]	0.5993 [0.5038, 0.6948]	0.9579	0.7985 [0.7189, 0.8688]	0.4146 [0.2666, 0.5641]
Macenko	0.8000 [0.7198, 0.8510]	0.4894 [0.3921, 0.5878]	0.9534	0.7806 [0.6930, 0.8585]	0.2195 [0.1000, 0.3500]
Vahadane	0.7862 [0.7167, 0.8433]	0.4405 [0.3454, 0.5378]	0.9389	0.7579 [0.6716, 0.8362]	0.1463 [0.0476, 0.2632]

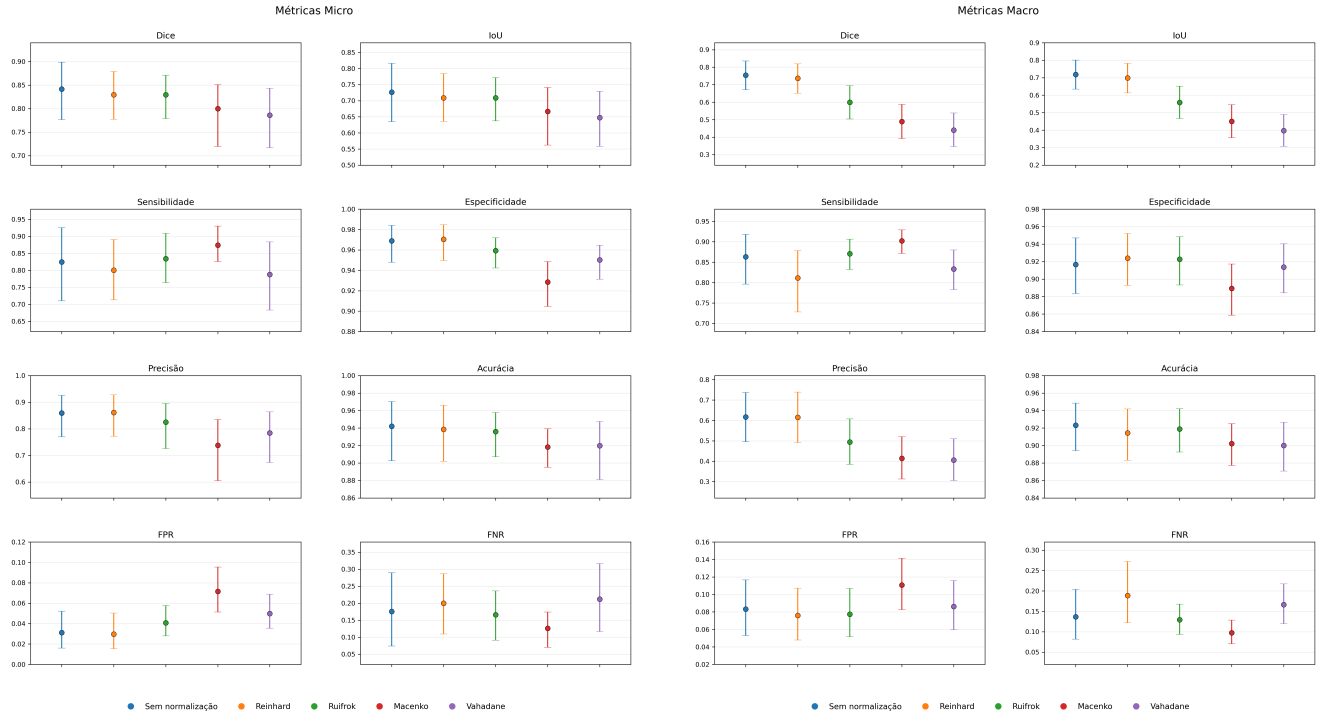


Figure 2: Micro- and macro-averaged DIAGSET test metrics across normalization strategies. Micro metrics pool pixel counts before metric computation, whereas macro metrics compute patient-level values before averaging. Error bars denote 95% patient-bootstrap confidence intervals where available.

Ruifrok obtained the highest AUROC point estimate and the highest Dice among positive patients, indicating a stronger probability ranking and favorable behavior on lesion-bearing cases. Thus, the most favorable normalization changes when the question moves from global segmentation overlap to ranking quality or positive-patient localization.

The patient-level decomposition exposes the main limitation and trade-off. Ruifrok, Macenko, and the non-normalized setting reached positive-patient Dice values near 0.78–0.80, indicating that the workflow could localize annotated lesion regions in many positive cases. However, stronger positive-patient behavior did not always coincide with conservative negative-patient behavior: the non-normalized and Reinhard settings suppressed all positive predictions in 30 of 41 negative patients, whereas Vahadane did so in only 6 of 41. Even the most conservative settings therefore left positive predictions in 11 negative test patients.

5. Discussion

The DIAGSET experiment supports three paper-level conclusions. First, the proposed workflow can compare pre-processing and normalization choices under a traceable, patient-level protocol. The comparison is not a loose collection of model runs: curation, split assignment, smart sampling, checkpoint selection, ensemble composition, threshold calibration, and final inference are connected through explicit artifacts and compatibility checks.

Second, color normalization did not produce a uniform improvement. No normalization was best under pixel-pooled and patient-averaged Dice, Ruifrok was best under AUROC and positive-patient Dice, and Macenko was best under DIAGSET-style patch AvAcc. This pattern agrees with the broader caution that stain preprocessing choices can change model behavior without guaranteeing a single dominant strategy across all metrics [25]. In this experiment, the choice of metric changes the apparent winner.

Table 2

DIAGSET-style patch recognition on the held-out test set. The primary comparison is before patient-level suppression; the last column reports AvAcc after applying the frozen patient-level suppression rule.

Normalization	Accuracy	AvAcc	Sensitivity	Specificity	Post-supp. AvAcc
None	0.9323	0.8923	0.8283	0.9562	0.8970
Reinhard	0.9296	0.8811	0.8036	0.9586	0.8854
Ruifrok	0.9325	0.8997	0.8474	0.9520	0.9005
Macenko	0.9166	0.9009	0.8758	0.9260	0.9015
Vahadane	0.9183	0.8715	0.7968	0.9462	0.8715

Third, patient-level false positives remain the primary limitation. High Dice in positive patients is not sufficient for a clinically conservative interpretation if negative patients often receive some positive prediction. The current results therefore support methodological viability and reproducible comparison, but not standalone clinical deployment. Improving calibration, patient-level post-processing, and false-positive suppression should be prioritized before extending claims beyond this experimental setting.

The study also has clear scope limits. DIAGSET is a single prostate histopathology dataset, and the workflow has not yet been validated here on other organs, scanner distributions, annotation protocols, or cancer types. The stored patient identifiers enforce split isolation within the repository contract, but they do not independently verify biological patient uniqueness outside the source metadata. Finally, the Vahadane condition corresponds to the fixed-source runtime approximation used in the experiment, not exact per-patch Vahadane fitting.

6. Conclusion

This paper presented a reproducibility-oriented workflow for whole-slide cancer segmentation and evaluated it on DIAGSET. The main contribution is methodological: the workflow treats the full experimental chain as the object of study, from slide ingestion and patch curation through patient-level splitting, training, validation-only ensemble optimization, and frozen test inference. On DIAGSET, the workflow produced interpretable metric-specific findings rather than a single universally superior normalization strategy. No normalization led pixel-pooled and patient-averaged Dice, Ruifrok led AUROC and positive-patient Dice, and Macenko led DIAGSET-style patch AvAcc. At the same time, persistent false positives in negative patients limit clinical interpretation. Future work should evaluate the same artifact-preserving protocol on external datasets and develop stronger patient-level calibration and false-positive-control mechanisms.

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